Riddha Manna, Ph.D.

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Academic qualifications:

- **Doctor of Science (Ph.D.) in Computational and Quantitative Biology (2021-2024)**
  Supervised by Dr. Ana Marija Jakšić and Prof. Kathryn Hess Bellwald, Brain Mind Institute, School of Life Sciences, **École Polytechnique Fédérale de Lausanne**, Lausanne, Switzerland
- **Master of Technology in Biomedical Engineering (2017-2019) (GPA 9.61/10)**
  *Department of Biosciences & Bioengineering, **Indian Institute of Technology Bombay**, Mumbai, India*
- **Bachelor of Technology in Biotechnology (2013-2017) (GPA 8.88/10**)
  *Department of Biotechnology, **National Institute of Technology Durgapur**, Durgapur, India*
- **Indian School Certificate Examination (2012) (91.5%)**
  Indian Certificate of Secondary Education Examination (2010) (94.6%)
  ***Calcutta Boys' School**, Kolkata, India*

Skills:

- **Programming, application development, and technical software:** Python, R, ABB RAPID, Julia, Java, C, C#, Perl, MATLAB, SQL, Processing, Android Studio, Swift, LabView, COMSOL Multiphysics, JavaScript, Git, Linux scripting, Robot Studio
- **Collaborative robotics:** Design, implementation and maintenance of automated protocols
- **Statistical analysis and Machine learning**
- **Mathematical modeling and simulations, Topological Data Analysis, Information Theory**
- **Electronic design, fabrication and firmware development**
- **Machine vision, image processing, object detection, classification and tracking, behavioral analysis**
- **Biotechnology:** Immunological, biochemical and molecular biology, microbiological assays and culturing techniques; Bioinformatics; ***Drosophila* handling and genetics**
- Microfluidic device fabrication using **photolithography (clean room work)**, **soft lithography** and **hot embossing**
- **2D and 3D CAD**
- **3D printing, machining, laser cutting & etching techniques**
- **Design thinking**, **business model development**, **survey design** (field experience)

Selected recent research experiences:

Unraveling behavioral individuality in *Drosophila melanogaster*: An automated high-throughput screening approach (January January 2021 – December 2024)

Ph.D. thesis project under Dr. Ana Marija Jakšić, at the Experimental Evolutionary Neurobiology Lab, BMI, SV, EPFL **<link>**

Developed a high throughput platform and tested ~6,000 flies from 90 isogenic lines in a cognitive assay, revealing that genetics, experience, and learning shape behavior. Also developed a robotic system that autonomously screens over 1,000 flies daily.

***Detection* of sickle cell disease at p-o-c using mobile phone microscopy** (July 2019 – September 2020)

Project Research Associate at Microfluidics & Biological Physics Group, Biosciences & Bioengineering, IIT Bombay

Developed a platform combining blood chemistry, microfluidics, mobile phone microscopy, and image processing for automated diagnosis of sickle cell disease at the point-of-care.

Towards development of an integrated system for diagnosis of infectious diseases of the blood (2018 – 2019)

M. Tech project under the guidance of Prof Debjani Paul (Department of Biosciences & Bioengineering, IIT Bombay) and Prof Kantimay Das Gupta (Department of Physics, IIT Bombay) **<link>**

Developed a microfluidic step-emulsification device to generate monodispersed droplets for encapsulation of single RBCs inside individual droplets for single-cell analysis. Developed a portable battery-operated single-channel fluorometer.

Publications and preprints:

- **Manna, R**., Štampar, T., Tomić, I. and Jakšić, A.M., 2025**. Robotic platform for extreme-throughput cognitive behaviour phenotyping in Drosophila.** (in preparation)
- **Manna, R**., Brea, J., Braga, G.V., Modirshanechi, A., Tomić, I. and Jakšić, A.M., 2024. **Learning is a fundamental source of behavioral individuality.** ***bioRxiv***. **<link>** (version 1: Behavioral individuality is a consequence of experience, genetics and learning. <link>)
- George, J.E., **Manna, R.,** Mange, J., Roy, S., Kumari, S., Raza, R. and Paul, D. 2025. **Rapid and high-throughput generation of agarose and gellan droplets by pump-free microfluidic step emulsification.** (submitted)
- D'Costa, C., Sharma, O., **Manna, R.**, Singh, M., Singh, S., Singh, S., Mahto, A., Govil, P., Satti, S., Mehendale, N., Italia, Y. and Paul, D. 2024. **Differential sensitivity to hypoxia enables shape-based classification of sickle cell disease and trait blood samples at point of care. *Bioengineering & Translational Medicine.* <link>**
- George, J.E., **Manna, R.,** Roy, S., Kumari, S. and Paul, D., 2023. **Pump-free and high-throughput generation of monodisperse hydrogel beads by microfluidic step emulsification for dLAMP-on-a-chip. *bioRxiv*. <link>**
- D'Costa, C., Sharma, O., **Manna, R.,** Singh, M., Samrat, S., Singh, S., Mahto, M., Govil, P., Satti, S., Mehendale, N., Italia, Y. and Paul, D. 2020. **Differential sensitivity to hypoxia enables shape-based classification of sickle cell disease and trait blood samples. medRxiv. <link>**
- Naik, P., **Manna, R.**, Paul D. 2019. **Nucleic Acid Amplification on Paper Substrates**. In: Bhattacharya S., Kumar S., Agarwal A. (eds) Paper Microfluidics. Advanced Functional Materials and Sensors. **Springer,** Singapore. **<link>**
- Pal, S., Kundu, A., Banerjee, T.D., Mohapatra, B., Roy, A., **Manna, R.**, Sar, P. and Kazy, S.K. 2017. **Genome analysis of crude oil degrading Franconibacter pulveris strain DJ34 revealed its genetic basis for hydrocarbon degradation and survival in oil contaminated environment. *Genomics*. <link>**

Patent contributions and commercialisations:

- Point of care sickle cell test. (Indian patent application filed: 2016, granted 2023), licensed to Lords Mark Ind. Pvt. Ltd. for field testing and commercialization as 'ShapeDx: A high-accuracy, point-of-care sickle cell test'.

Awards and fellowships:

- **MHRD GATE Fellowship** (2017 – 2019) securing National rank **87 (98.87 percentile)**.
- **Tata Fellowship** (2017 – 2019) by the Tata Center for Technology and Design, IIT Bombay for designing technology solutions for resource-constrained communities and social entrepreneurship.
- **Dr. P. K. Patwardhan Technology Development Award, IIT Bombay** (2022) in recognition of innovative R&D work on 'ShapeDx: A high-accuracy, point-of-care sickle cell test'.

Selected conference contributions:

- **The Allied Genetics Conference 2024** (2024; Washington D.C., USA) *"Chance and genetics shape the diversity of individual cognitive behavior in Drosophila melanogaster"* (oral presentation)
- **Swiss Drosophila Meeting** (2024; Lausanne, Switzerland) *"Interplay of genotype and experience shapes cognitive individuality in Drosophila melanogaster"* **(Awarded best poster)**

- **The Annual Meeting of the NeuroLeman Network and Doctoral Schools** (2024; Villars-sur-Ollon, Switzerland) *"Individuality and genetics of cognitive behaviour in Drosophila melanogaster revealed by visual learning"* (poster)
- **64th Annual Drosophila Research Conference** (2023; Chicago, IL, USA) *"Automated cobot-assisted high-throughput phenotyping of cognition in behaving Drosophila melanogaster"* (oral presentation)
- **Swiss Drosophila Meeting** (2023; Fribourg, Switzerland) *"Automated high-throughput phenotyping of cognition in behaving individual Drosophila melanogaster reveals genetically influenced degree of individuality in learning"* **(Special mention: oral presentation)**
- **Biosciences & Bioengineering Seminar, IIT Bombay** (2023; Mumbai, India) *"Automated cobot-assisted high-throughput phenotyping of cognition in behaving Drosophila melanogaster reveals genetic control of cognition and cognitive individuality"* (oral presentation)
- **3rd EPFL Symposium on Modelling Sensorimotor Functions** (2023; Lausanne, Switzerland) *"Individuality and genetics of cognitive behaviour in Drosophila melanogaster revealed by visual learning"* (oral presentation)
- **28th International Conference on Miniaturized Systems for Chemistry and Life Sciences (MicroTAS)** (2024; Montréal, QC, Canada) "*High-throughput generation of spherical and disk-shaped droplets and hydrogel beads for biomedical applications using microfluidic step emulsification"* (poster)
- **Global Health Workshop** (2023; Mumbai, India) *"Point-of-care detection of sikle cell disease and trait, through microscopic determination of red blood cell shapes induced by differential hypoxia"* (**LabOnAChip poster award)**
- **26th International Conference on Miniaturized Systems for Chemistry and Life Sciences (MicroTAS)** (2022; Hangzhou, China) "*Pump-free generation of hydrogel beads by microfluidic step emulsification"* (poster)
- **24th International Conference on Miniaturized Systems for Chemistry and Life Sciences (MicroTAS)** (2020; Online) "*Using RBC shapes to distinguish between sickle cell disease and trait samples"* (poster)
- **SelectBio Microfluidics & LAB-ON-A-CHIP Conference** (2019; Mumbai, India) *"A microfluidic step-emulsification device for single cell encapsulation in droplets"* **(Awarded best poster)**

Responsibilities:

- Vice President, YUVA Indian Students' Association at EPFL-UNIL (2023-2024)
- Event Manager, PrangaN@Swiss, non-profit organization of Bengali diaspora in Switzerland (2021-2024)
- Marketing Head, Symbiotek Council, Department of Biosciences & Bioengineering, IIT Bombay (2019)
- Head, RoboCell, Center for Cognitive Activities, NIT Durgapur (2016 – 2017)
- Invited exhibitor and Volunteer, EPFL Open Doors, EPFL (2022)
- Organizing UK-India Newton-Bhabha Fund RSC Researchers link Workshop 2019, Tata Symposium 2018 & 2019, M. Tech admission process of Biosciences & Bioengineering IIT Bombay (2019)
- Arranged eight fests and taught at ten workshops (200+ participants each) for manual and autonomous robotics as a member of the RoboCell, Center for Cognitive Activities, NIT Durgapur (2013 – 2017)

Workshops and certifications:

- Workshop: **"Physics of Behavior",** 2024, EPFL, Lausanne, Switzerland
- Workshop: "**Experimental Evolution: Bringing theory and practice together"**, 2022, Vienna Graduate School of Population Genetics, Institut für Populationsgenetik, Vetmeduni, Vienna, Austria
- Workshop: **"Image data science with Python and Napari",** 2022, EPFL, Lausanne, Switzerland
- Workshop: **"Systems Biology of the Brain"**, 2021, University of Fribourg, Fribourg, Switzerland
- Introductory Course on **Population Genetics**, 2020, Vienna Graduate School of Population Genetics, Institut für Populationsgenetik, Vetmeduni, Vienna, Austria
- Short term course on **Recent Trends in Industrial Biotechnology** (RITB-2016), November 2016, Department of Biotechnology, NIT Durgapur, India
- **Green Revolution Certification Program**, International Center for Culture & Education to Educate, Inspire & Act Against Climate Change, supported by United Nations Framework Convention on Climate Change

Teaching and mentoring:

- Teaching Assistant for the course of **Genomics and Bioinformatics, School of Life Sciences, EPFL** (2023)
- Teaching Assistant for the course of **Integrated laboratory in Life Sciences I & II, School of Life Sciences, EPFL** (2023)
- Teaching Assistant for the course of **Product Design and Systems Engineering, Institute of Electrical and Micro Engineering, EPFL** (2022)
- Teaching Assistant for the course of **Experimental Techniques in Biomedical Engineering (Microfluidics module), Biosciences & Bioengineering, IIT Bombay** (2018 – 2019)
- Mentored four Bachelor's and two Master's students in the lab during Ph.D. project

Memberships:

Genetics Society of America, Swiss Society for Neuroscience, Society for the Study of Evolution

Other research experience and projects:

Genome analysis of crude oil degrading *Franconibacter* sp. DJ34 (2016 – 2017)
B. Tech project under the guidance of Prof Kazy Sufia Khannam (Department of Biotechnology, NIT Durgapur) **<link>**
Identified membrane transport proteins and aromatic compounds degradation pathways in the organism from its genome sequence to assess its usability in bioremediation of crude oil. Performed a comparative genomic analysis with other known crude oil-degrading bacteria.

Development of methodology for mass spectrometry-based imaging (Summer internship, 2016)
Guide: Prof Soumen Kanti Manna (Biophysics & Structural Genomics Division, Saha Institute of Nuclear Physics, Kolkata, India) **<link>**
Developed a novel algorithm to image mass spectra (MALDI-TOF MS) from spatially distributed spots using R and MATLAB.

Biomarkers and their predictive value in measuring malnutrition (Summer internship, 2015)
Guide: Prof Ranjan Kumar Nandy (Division of Bacteriology, National Institute of Cholera and Enteric Diseases, Kolkata, India) **<link>**
Detected and quantified biomarkers of tropical enteropathy in clinical samples using ELISA and HPLC, and analyzed their correlation with anthropometric z-scores of the subjects.

Wireless insole plantar-pressure mapping system for diagnosis of diabetic-feet (2019) *Medical instrumentation course project under the guidance of Prof Soumya Mukherji (Department of Biosciences & Bioengineering, IIT Bombay)*
Independently developed a novel technology, its instrumentation, and mobile (Android and iOS) and PC/MAC (java based) applications for high-resolution pressure mapping using a cheap commercially available piezoresistive polymer.

Association between Vimentin and Tubulin contributes to Paclitaxel resistance in MCF-7 Human breast cancer cell lines (2019)
Cell Mechanics and Mechanobiology course white paper project under the guidance of Prof Shamik Sen (Department of Biosciences & Bioengineering, IIT Bombay)

Affordable pressure mat for wheelchairs for diagnosis of pressure ulcers (2018) *Technology Design and Development Laboratory course project under the guidance of Prof Santosh Noronha and Prof Arindam Sarkar (Department of Chemical Engineering, IIT Bombay)*

Microfluidics for Education (2018) *Seminar white paper project under the guidance of Prof Debjani Paul (Department of Biosciences & Bioengineering, IIT Bombay)*
Explored the possibility of using microfluidic technologies for high-school level practical science experiments.

A Programmable Microfluidic Multiplexer Device for the diagnosis of oral cancer (2018) *Microfluidics: Physics and Applications course white paper project under the guidance of Prof Debjani Paul (Department of Biosciences & Bioengineering, IIT Bombay)*

Automated parallel pumping system to analyze and compare the performance of commercial domestic water purifiers (2018)
Guide: *Prof Murali Sastry, Chief Executive Officer, IIT Bombay-Monash Research Academy*

Grainbuoy: A cheap flood-resistant silo (2017) *White-paper project and presentation for "Startup alley," a B-Plan competition at Entrepreneurship Fair 2017, organized by the National Entrepreneurship Network, NIT Durgapur, with the theme of rural development.* ***(Awarded best project)***

Selected pet projects (independent) (2010 – Present) **<link>** **<link>**
Non-invasive sleep paralysis detection and intervention system, Facebook message-controlled robot and home automation, pocket-sized oscilloscope, pulse-rate monitor, mini spectrophotometer, maze-solving line-follower, automatic fire extinguisher, all-terrain rover (caterpillar and shrimp mechanism), thermocycler, persistence of vision binary clock, solar panel alignment system.

Other conference attendance and contributions:

- **Physics of Living Systems Seminar** (2023; Lausanne, Switzerland) "Automated cobot-assisted high-throughput phenotyping of cognition in behaving *Drosophila melanogaster"* (oral presentation)
- **Super Sandwich Seminar** (2023; Lausanne, Switzerland) *"Automated cobot-assisted high-throughput phenotyping of cognition in behaving Drosophila melanogaster"* (oral presentation)
- **Super Sandwich Seminar** (2022; Lausanne, Switzerland) *"Development of an automated high-throughput system for phenotyping cognition in behaving Drosophila melanogaster"* (oral presentation)
- **Brain Mind Institute Seminar** (2021; Lausanne, Switzerland) *"Artificial selection for cognitive ability in Drosophila melanogaster"* (oral presentation)
- **EPFL EDCB (Doctoral Program in Computational and Quantitative Biology) Day** (2022; Lausanne, Switzerland) *"Automated high-throughput phenotyping of cognition in behaving Drosophila melanogaster for long term evolutionary experiments"* (poster)
- **The Asia-Pacific Conclave on Engineering and Healthcar**e (2024; Mysuru, India) *"Optimization of a microfluidic device for generation of hydrogel beads by step emulsification"* (poster)
- **Global Health Workshop** (2023; Mumbai, India) *"High-throughput generation of monodisperse gellan hydrogel beads using microfluidic step emulsification"* (poster)
- **8th Annual Meeting of Proteomics Society, India (PSI)** in conjunction with the **3rd Meeting of Asia Oceania Agricultural Proteomics Organization (AOAPO)** and **International Conference on Functional and Interaction Proteomics: Application in Food and Health** (2016; New Delhi, India) *"Evaluation of MALDI-MS Platform for Peptidomic Lipidomic and Metabolomic Imaging Studies in the Context of Non-alcoholic Fatty Liver Disease"* (poster)
- **UK-India Newton-Bhabha Fund RSC Researchers link Workshop: Microfluidics for Anti-microbial Resistance**, IIT Bombay, 2019
- **Tata Center for Technology and Design Symposium, IIT Bombay,** 2018 & 2019.
- **Tata Center for Technology and Design Symposium, Massachusetts Institute of Technology**, 2018
- **Wadhwani Research Center for Bioengineering Industry Day, IIT Bombay**, 2018

Hobbies:

Photography, competitive coding, playing football, motorcycling.

References:

Dr. Ana Marija Jakšić
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Prof. Kathryn Hess Bellwald
Full Prefessor, Laboratory of Topology and Neuroscience
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